

On coordinate frames relevant for pulsar physics

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Abstract

Pulsars can be either isolated or binary (or even in triple) systems. Observational features of pulsars are used to probe various aspects of fundamental physics, including emission mechanism, gravitational physics, etc. Theoretical models of various phenomena need different coordinate frames. Often different physical processes affect each other and we need to use multiple coordinate frames and relations between those. This short review presents an extensive set of coordinate frames and relations between those.

Keywords:

pulsars, spin-frequency, dynamics

1. Introduction

The investigation of pulsars has significantly advanced our understanding of several important areas in astrophysics including the internal physics of neutron stars and the behavior of matter under extreme physical conditions (Basu et al., 2025). As pulsars act as remarkably precise cosmic clocks, with some millisecond pulsars exhibiting timing stability comparable to that of atomic clocks, these are also useful for testing general relativity and other theories of gravity (Bagchi & Torres, 2014), studying the properties of the interstellar medium, constructing a relativistic deep-space positioning system (Bunandar et al., 2011), and searching for low-frequency gravitational waves (Agazie et al., 2024).

Tests of gravity are possible through pulsar timing analysis, which involves accurately calculating the rotational phases of the pulsar by modeling the delays between successive Times of Arrival (ToAs) of the pulses. There are various types of delays that the signal from a pulsar experiences, regardless of whether the pulsar is isolated or part of a binary system, e.g., dispersion delay, the solar system Römer delay, the solar system Shapiro delay, etc. The signals from binary pulsars experience some additional delays mostly due to the orbital motion and the extra curvature of the spacetime around the companion. Some of these binary specific delays are the binary Römer delay, the binary Shapiro delay, the binary Einstein delay, delays due to light-bending effects, etc (Lorimer & Kramer, 2004).

Nearly ten percent of the presently known radio pulsars are members of binary systems. Binaries in which the companion of the pulsar is also a compact object, such as a white dwarf, another neutron star, or a stellar-mass black hole, provide unique opportunities to test theories of gravity, including general relativity, scalar-tensor theory, and tensor-vector-scalar theory (Stairs, 2003).

In most of the analytical studies aiming to model the emission physics or the gravitational physics or the effect of gravitation on the pulsar signal, one needs to use more than one coordinate frames. In this short review, we provide details of some of such frames that are commonly used. In Section 2, we define these frames and in Section 3, we present the transformation relations between these frames. Most of these frames have been used recently in studies of the effect of the light bending phenomenon on the signal of a pulsar in a binary, due to the gravitational field of the companion (Debnath et al., 2023; Debnath & Bagchi, 2025).

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Preprint submitted to New Astronomy

August 28, 2026

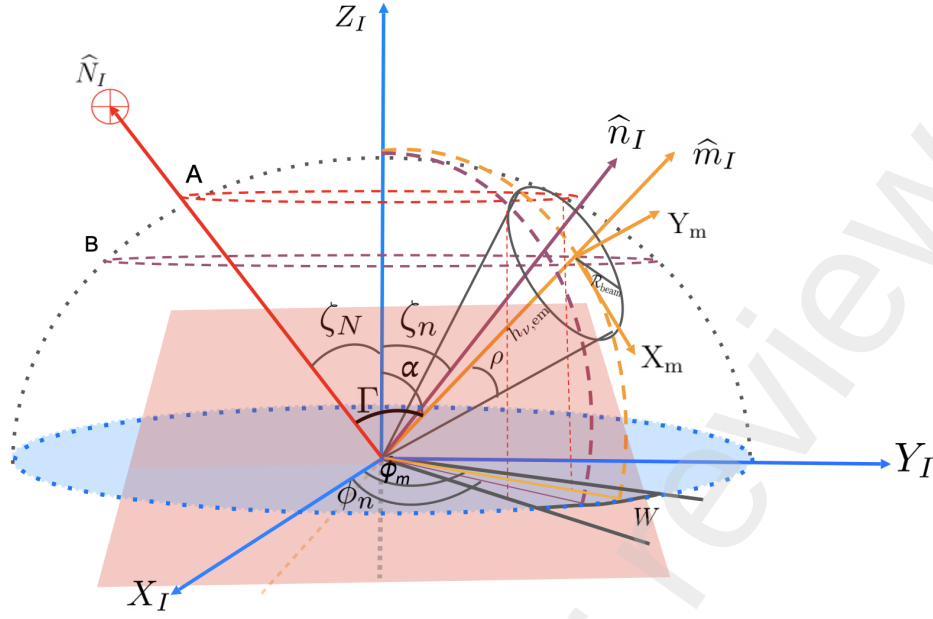


Figure 1: The beam of a pulsar and the Line-of-Sight (LoS). The meanings of various vectors and angles have been explained in the text. The red plane is perpendicular to the LoS, going through the pulsar, i.e., the sky-plane, and the blue plane is the $X_I Y_I$ plane. Note that here we have assumed that the pulsar rotates clockwise if viewed by the observer. If the case is reversed, then the Z_I -axis would be the negative direction of the spin axis. We always take $\hat{m}_I$ along the side of the magnetic axis that is closer to the Z_I -axis.

2. Some important frames in pulsar physics

Various reference frames commonly used in pulsar physics can be broadly classified into two classes, e.g., (i) frames needed to study isolated as well as binary pulsars and (ii) frames needed to study phenomena specific to binary pulsars.

2.1. Frames needed to study isolated as well as binary pulsars

Before discussing any frame, we first present the geometry of the pulsar beam. We know that a neutron star is viewed as a pulsar when its magnetic axis is misaligned from the spin axis and as the pulsar rotates, its magnetic axis also rotates around the spin axis (Radhakrishnan & Cooke, 1969). Fig. 1 shows a schematic diagram of the emission geometry of a pulsar at an arbitrary time t . In this figure, we have shown different frames that are useful to study isolated pulsars, as we discuss next. We will define various angles while discussing these frames that are needed to understand the pulsar beam model in detail as discussed in Subsection 2.1.3.

2.1.1. The pulsar frame (I-frame)

We define the ‘pulsar frame’ (or the I-frame) with its origin at the center of the pulsar. The Z-axis of this frame (the Z_I -axis) is aligned with the spin axis of the pulsar. The line-of-sight (LoS) vector in this frame is defined as the direction from the center of the pulsar to the observer. The X-axis of the I-frame (the X_I -axis) is chosen to lie in the plane formed by the spin axis and the LoS vector. The unit vector along the LoS is denoted by $\hat{N}_I$.

The angle between the spin axis and the LoS is denoted by ζ_N , so $\hat{N}_I$ can be written as:

$$\hat{N}_I = [\sin \zeta_N, 0, \cos \zeta_N]. \quad (1)$$

The direction of the Y-axis of the I-frame (the Y_I -axis) is then determined by the right-handed cross product:

$$\hat{y}_I = \hat{z}_I \times \hat{x}_I. \quad (2)$$

where $\hat{y}_I$, $\hat{z}_I$, and $\hat{x}_I$ are the unit vectors along the X_I , Y_I , and Z_I axes, respectively.

Without any loss of generality, we also choose the magnetic axis to lie in the $X_I Z_I$ plane at time $t = 0$. Then, the rotational phase of the magnetic axis (or pulsar phase) becomes:

$$\phi_m = \frac{2\pi}{P_s} t, \quad (3)$$

where P_s is the spin period of the pulsar. Thus, ϕ_m is the angle between the X_I -axis and the projection of the magnetic axis onto the $X_I Y_I$ plane.

The angle between the spin axis (Z_I) and the magnetic axis is denoted by α . At any arbitrary time t , the unit vector along the magnetic axis $\widehat{m}_I$ can be written as:

$$\widehat{m}_I = [\sin \alpha \cos \phi_m, \sin \alpha \sin \phi_m, \cos \alpha]. \quad (4)$$

The angle between the magnetic axis and the LoS is Γ , which varies with the rotational phase (ϕ_m) of the pulsar as the magnetic axis rotates about the spin axis. From Eqs. (1) and (4) we get,

$$\Gamma = \cos^{-1} (\sin \alpha \cos \phi_m \sin \zeta_N + \cos \alpha \cos \zeta_N) \quad (5)$$

For a particular pulsar, among α , ϕ_m , and ζ_N , only ϕ_m is time dependent. From Eq. (5), one can show that Γ is minimum when $\phi_m = 0$ and maximum when $\phi_m = \pi$. The minimum value of Γ (at $\phi_m = 0$), known as the angle of closest approach (β , not shown in the figure), can be written as:

$$\beta = \zeta_N - \alpha. \quad (6)$$

As shown in Fig. 1, $\widehat{n}_I$ is the unit vector along a generic light ray, which is considered to make an angle ζ_n with the Z_I -axis, while ϕ_n denotes the angle made by its projection onto the $X_I Y_I$ plane with the X_I -axis. Then, $\widehat{n}_I$ can be written as:

$$\widehat{n}_I = [\sin \zeta_n \cos \phi_n, \sin \zeta_n \sin \phi_n, \cos \zeta_n]. \quad (7)$$

This frame is particularly useful for studying various aspects of pulsar physics, such as the structure of the pulsar and its emission mechanism.

2.1.2. The beam frame or magnetic axis frame (m-frame)

We define the ‘beam frame’ at the center of the beam cross-section, located at the position where the emission originates, i.e., at $h_{v,em}$. The Z -axis of the beam frame (the Z_m -axis) is aligned with the magnetic axis of the pulsar. The X and the Y axes of this frame (the X_m and the Y_m axes, respectively) span the plane perpendicular to Z_m , with the beam cross-section lying in the $X_m Y_m$ plane. The X_m -axis is chosen to lie in the plane formed by Z_I -axis and Z_m -axis. The direction of the Y_m -axis is then determined by the right-handed cross product:

$$\widehat{y}_m = \widehat{z}_m \times \widehat{x}_m, \quad (8)$$

Here, $\widehat{x}_m$, $\widehat{y}_m$, and $\widehat{z}_m$ denote the unit vectors directed along the X_m , Y_m , and Z_m axes, respectively. Hence, $\widehat{z}_m$ and $\widehat{m}_I$ are equivalent. This frame is particularly useful for studying properties associated with the emission mechanism, e.g., the beam geometry, internal structure of the emission cone, etc.

As shown in Fig. 1, as the beam rotates, the points of the beam that traverse the path marked ‘A’ would fall along the LoS and would be visible. The magnetic axis traverses along the path ‘B’ which is around the LoS and does not fall on it.

2.1.3. Some parameters of a simple beam model

We now try to understand a simple emission model with the help of various angles defined in earlier subsections. Many pulsar emission phenomena can be modelled with this simple emission model proposed by Radhakrishnan & Cooke (1969). According to this model, a pulsar emits radiation along a conal beam centered on its magnetic axis. The beam is generated at an emission height that depends on the observing frequency, given by Lorimer & Kramer (2004) as:

$$h_{v,em} = 400 \text{ km} \left(\frac{\nu}{10^9 \text{ Hz}} \right)^{-0.26} \left(\frac{\dot{P}_s}{10^{-15} \text{ s s}^{-1}} \right)^{0.07} \left(\frac{P_s}{1 \text{ s}} \right)^{0.30}, \quad (9)$$

where $\dot{P}_s$ is the time derivative of the pulsar spin period (P_s) and ν is the observing frequency.

Following Gil, Kijak & Seiradakis (1993), the half-opening angle of the beam, ρ , is related to the observed pulse width W , the magnetic inclination α , and the angle β , as given in Eq. (6) as:

$$\sin^2\left(\frac{\rho}{2}\right) = \sin^2\left(\frac{W}{4}\right) \sin \alpha \sin \zeta_N + \sin^2\left(\frac{\beta}{2}\right). \quad (10)$$

Here, W is the observed pulse width, which can be defined as:

$$W = \frac{\Delta t}{P_s} \times 2\pi, \quad (11)$$

where Δt is the total time the LoS remains inside the beam. In other words, Δt is the total time during one rotation for which the angle between the magnetic axis and the LoS, i.e., Γ , remains less than the half-opening angle ρ of the cone.

2.2. Frames needed to study physics relevant for binary pulsars

When a pulsar is in a binary system, it moves in an elliptical orbit around the barycenter (center of mass) of the system. In such a case, we often need to use some additional frames, some of these have the barycenter as the origin and some the center of the companion as the origin. We might even need to define frames centered at the center of the pulsar.

In case of a binary pulsar, the definition of LoS is different from that for an isolated pulsar. For a binary pulsar (or any binary stellar system) the LoS is defined from the barycenter to the observer¹.

The plane in which a pulsar and its companion move is known as the ‘orbital plane’. In Keplerian theory, this is a well defined plane. However, in the strong gravitational regime, when general relativistic effects are pronounced, the motions become more complicated, which is usually modeled as the change in the orientation of the orbital plane.

One more important plane is the sky-plane’. In case of a binary pulsar, the sky-plane is the plane containing the barycenter and perpendicular to the LoS (shown as the red plane in Fig. 2). For an isolated pulsar, the sky-plane can be taken simply as the plane perpendicular to the LoS and passing through the center of the pulsar (not shown in any of the figures).

2.2.1. The sky frame (s-frame)

To define the sky-frame or the s-frame, we first denote the unit vector along the LoS for a binary pulsar is $\hat{N}_s$ (in this s-frame). The intersection of the sky-plane with the orbital plane of the binary is known as the ‘line of node’. The point of intersection of the orbital ellipse where the orbiting object (here the pulsar) moves away from the observer is called the ‘ascending node’ (AN) and the other point of intersection where the orbiting object moves toward the observer is known as the ‘descending node’ (DN).

Then, the s-frame is defined as the frame having origin at the barycenter and the X-axis and Y-axis are on the sky-plane, i.e., the sky plane can be called as the $X_s Y_s$ plane. The Z-axis of the s-frame (the Z_s -axis) is directed opposite to $\hat{N}_s$. The X-axis (X_s) is chosen to lie along the line joining the barycenter and the ascending node (AN) of the orbit. The Y-axis (Y_s) is then determined using the right-handed cross product:

$$\hat{y}_s = \hat{z}_s \times \hat{x}_s, \quad (12)$$

where $\hat{x}_s$, $\hat{y}_s$, and $\hat{z}_s$ denote the unit vectors directed along the X_s , Y_s , and Z_s axes, respectively. This frame is often used in studies of orbital dynamics of pulsars and associated observational features.

¹The definition for LoS for a single pulsar is different, conventionally, it is defined from the pulsar to the observer. However, the LoS for a single pulsar and the LoS for a binary pulsar are practically identical because, in any realistic scenario, the observer’s distance is much larger than the orbital size.

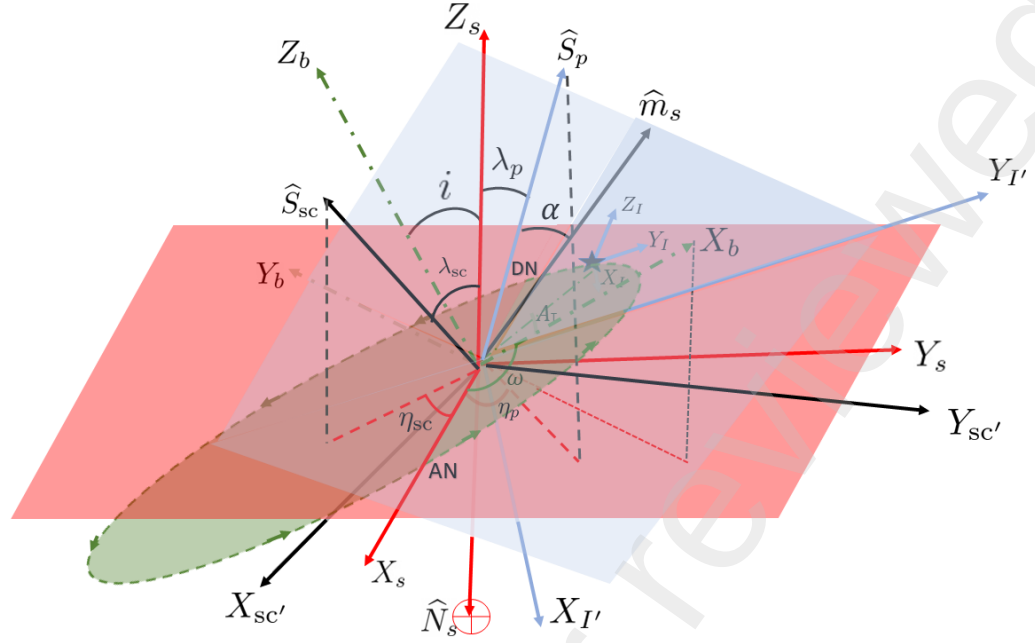


Figure 2: The orbital geometry of a pulsar (marked by a $\star$) in a binary system. The barycenter is taken as the origin of various frames shown. Details of the axes and angles can be found in the text. The sky plane (the $X_s Y_s$ plane) is shown with a light red color and the orbital plane (the $X_b Y_b$ plane) is shown with a green color. Direction of motion of the pulsar on the orbital ellipse has been shown with green arrows on the ellipse. The light blue plane is the $X'_I Y'_I$ plane that is parallel to the $X_I Y_I$ plane defined in Subsection 3.4. The unit vector along the spin axis of the pulsar is denoted by $\hat{S}_p$, which is physically along the Z_I -axis, but here shown along the Z'_I -axis, i.e., has been shifted parallelly to the barycenter. Similarly, the unit vector along the spin axis of the companion is denoted by $\hat{S}_{sc}$, which is physically along the Z_I -axis, but here shown along the Z'_I -axis, i.e., has been shifted parallelly to the barycenter. The ascending node has been denoted by AN and the descending node by DN. The direction of the earth is denoted by $\oplus$.

2.2.2. The sky frame shifted to the center of the pulsar (pS-frame) or to the center of the companion (cS-frame)

Often, the vectors that are defined with respect to the center of the pulsar, need to be expressed in the sky-frame. For this purpose, one can simply shift the origin of the sky-frame to the center of the pulsar without changing the direction of the axes (translation without any rotation). This frame can be called the pS-frame whose X , Y , and Z axes are denoted by X_{pS} , Y_{pS} , and Z_{pS} , respectively.

Similarly, one can shift the origin of the sky-frame to the center of the companion without changing the direction of the axes (translation without any rotation). This frame can be called the cS-frame whose X , Y , and Z axes are denoted by X_{cS} , Y_{cS} , and Z_{cS} , respectively.

Lorimer & Kramer (2004) expressed the orientation of the spin axis (the Z_I -axis) of the pulsar in the pS-frame as follows. The angle between the Z_{pS} -axis and the Z_I -axis is denoted by λ_p , and the angle between the X_{pS} -axis and the projection of the Z_I axis on the $X_{pS} Y_{pS}$ -plane is denoted by η_p . Similarly, one can express the orientation of the spin axis of the companion in the cS-frame. The angle between the Z_{cS} -axis and the Z_{sc} -axis is denoted by λ_{sc} , and the angle between the X_{cS} -axis and the projection of the Z_{sc} axis on the $X_{cS} Y_{cS}$ -plane is denoted by η_{sc} .

In the left panel of Fig. 3, we show the orientation of the I-frame with respect to the pS-frame. As the pS-frame is parallel to the s-frame, I-frame is oriented exactly the same way with respect to the s-frame as depicted in Fig. 2. Similarly, In the right panel of Fig. 3, we show the orientation of the sc-frame with respect to the cS-frame. As the cS-frame is parallel to the s-frame, sc-frame is oriented exactly the same way with respect to the s-frame as depicted in Fig. 2.

2.2.3. The orbital frame (b-frame)

We define the orbital frame with origin at the barycenter. The Z_b -axis is defined to be perpendicular to the orbital plane and pointed along the orbital angular momentum of the pulsar. In Fig. 2, the orbital motion of the pulsar is clockwise if viewed from the observer, hence, the

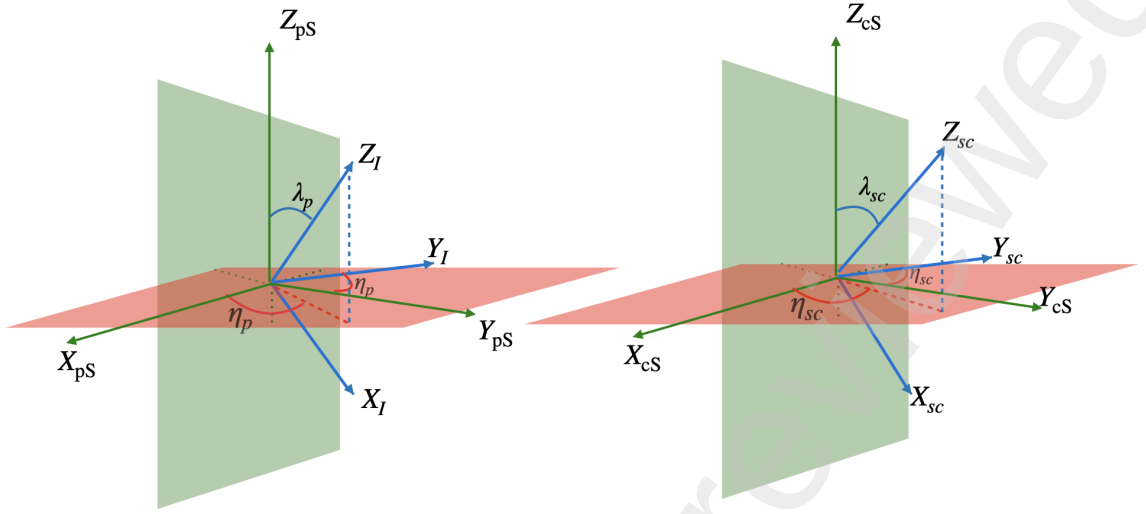


Figure 3: The left panel shows the I-frame and the pS-frame and the right panel shows the sc-frame and the cS-frame. These frames and various angles have been explained in the text.

Z_b -axis and LoS are on the opposite side of the orbital plane. If the orbital motion was taken anti-clockwise if viewed from the observer, the Z_b -axis and LoS would be of the same side of the orbital plane. The X_b -axis lies along the line connecting the barycenter to the initial periastron of the orbit². Then, the Y_b -axis is then obtained from the cross product:

$$\widehat{y}_b = \widehat{z}_b \times \widehat{x}_b. \quad (13)$$

Here, $\widehat{x}_b$, $\widehat{y}_b$, and $\widehat{z}_b$ denote the unit vectors directed along the X_b , Y_b , and Z_b axes, respectively.

This construction provides a well-defined right-handed coordinate system, referred to as the ‘orbital frame’ (or the b-frame), which is used to study binary pulsar motion and to test predictions of general relativity.

The inclination angle of the orbit is defined as the angle between Z_s -axis and Z_b and is denoted by i , and the angle between the X_s -axis and the X_b -axis is known as the longitude of the periastron and is denoted by ω . These angles are also shown in Fig. 2.

The above choice also implies that the angle between the $X_s Y_s$ plane and the $X_b Y_b$ plane, i.e., the angle between the sky-plane and the orbital plane is i , and this is often stated as the definition of the inclination angle.

2.2.4. The spinning-companion frame or the sc-frame

It is evident that the companion in a binary pulsar system can itself be a spinning object, such as a rotating white dwarf, another pulsar, or a spinning black hole. In such cases, certain calculations, particularly geodesic solutions around the spinning companion, i.e., in Kerr spacetime, are most conveniently performed in a coordinate system centered on the spinning companion, named as the ‘spinning-companion frame’ or the ‘sc-frame’.

The Z -axis of this frame denoted by Z_{sc} , is chosen to align with the spin axis of the companion. The X -axis of the sc-frame (or the X_{sc} -axis), is chosen in such a way that the $Z_{sc} X_{sc}$ plane contains the LoS translated to the center of the companion. The direction of the Y -axis of the sc-frame (the Y_{sc} -axis) is then determined by the cross product

$$\widehat{y}_{sc} = \widehat{z}_{sc} \times \widehat{x}_{sc}, \quad (14)$$

²Due to general relativistic effects, the orbital ellipse precesses in the orbit, one can model this periastron precession with respect to the X_b -axis.

where $\widehat{x}_{sc}$, $\widehat{y}_{sc}$, and $\widehat{z}_{sc}$ denote the unit vectors directed along the X_{sc} , Y_{sc} , and Z_{sc} axes, respectively.

2.2.5. The barycentric pulsar frame (I' -frame)

The I' -frame is defined with its origin at the center of the pulsar. We shift this origin to the barycenter keeping the directions of the axes unchanged. This gives a new frame called the I' -frame with its X , Y , Z axes labeled as $X_{I'}$, $Y_{I'}$, and $Z_{I'}$, respectively. As $Z_{I'}$ is taken along the spin axis of the pulsar, we can also say the spin vector of the pulsar is along the $Z_{I'}$ -axis.

As both the transformations, namely, the I' -frame from the I -frame and the s -frame from the pS -frame (Subsection 2.2.2) involve only translations without rotation, following Subsection 2.2.2, we can say that the angle between the Z_s -axis and the $Z_{I'}$ -axis is λ_p , and the angle between the X_s -axis and the projection of the $Z_{I'}$ axis on the $X_s Y_s$ -plane is η_p . These angles have been shown in Fig. 2.

2.2.6. The barycentric spinning-companion frame or the sc' -frame

The sc -frame is defined with its origin at the center of the pulsar. We shift this origin to the barycenter keeping the directions of the axes unchanged. This gives a new frame called the sc' -frame with its X , Y , Z axes labeled as $X_{sc'}$, $Y_{sc'}$, and $Z_{sc'}$, respectively. As the $Z_{sc'}$ -axis is taken along the spin axis of the companion, we can also say that the spin vector of the companion is along the $Z_{sc'}$ -axis. Hence, the unit vector along the $Z_{sc'}$ -axis, i.e., $\widehat{z}_{sc'}$ is equivalent to the unit spin vector of the companion ($\widehat{S}_{sc}$) as shown in Fig. 2.

As both the transformations, namely, the sc' -frame from the sc -frame and the s -frame from the cS -frame (Subsection 2.2.2) involve only translations without rotation, following Subsection 2.2.2, we can say that the angle between the Z_s -axis and the $Z_{sc'}$ -axis is λ_{sc} , and the angle between the X_s -axis and the projection of the $Z_{sc'}$ axis on the $X_s Y_s$ -plane is by η_{sc} . These angles have been shown in Fig. 2.

2.2.7. The p -frame and the c -frame

Two additional frames, not shown in Fig. 2, are often useful. Those are defined at the centers of the pulsar and its companion: the $X_p Y_p Z_p$ frame (the ' p -frame') and the $X_c Y_c Z_c$ frame (the ' c -frame'), respectively.

These frames are related to the b -frame through parallel translations along the line joining the pulsar, the barycenter, and the companion. Because the b -frame, the p -frame, and the c -frame are parallel to one another, any direction vector calculated in one frame will be identical in all three. Moreover, the direction of the LoS is effectively the same in all three frames, as in any realistic scenario, the observer's distance is much larger than the orbital size.

2.2.8. The T -frame

In certain calculations involving binary pulsars, the initial direction of the emitted light rays plays a significant role. For example, the bending of light in the gravitational field of the companion depends sensitively on the initial propagation direction of the rays (Debnath et al., 2023). To handle such cases, we construct the ' T -frame' for each of the light rays, depending on the initial direction of the light ray, but centered at the center of the companion (see Fig. 4).

The unit vector along the Z -axis of the T -frame or the Z_T -axis, denoted by $\widehat{z}_T$, is defined as:

$$\widehat{z}_T = \widehat{r}_b \times \widehat{n}_b, \quad (15)$$

where $\widehat{r}_b$ is the unit position vector of the center of the pulsar in the b -frame, and $\widehat{n}_b$ is the unit vector along the initial direction of the light ray from the pulsar (origin of the m -frame), also expressed in the orbital frame. The Y -axis of the T -frame is chosen to lie along the direction of $\widehat{r}_b$, i.e., the unit vector along the Y_T -axis is

$$\widehat{y}_T = \widehat{r}_b. \quad (16)$$

Finally, the unit vector along the X -axis of the T -frame (X_T) is determined by the cross product of $\widehat{z}_T$ and $\widehat{y}_T$,

$$\widehat{x}_T = \widehat{z}_T \times \widehat{y}_T. \quad (17)$$

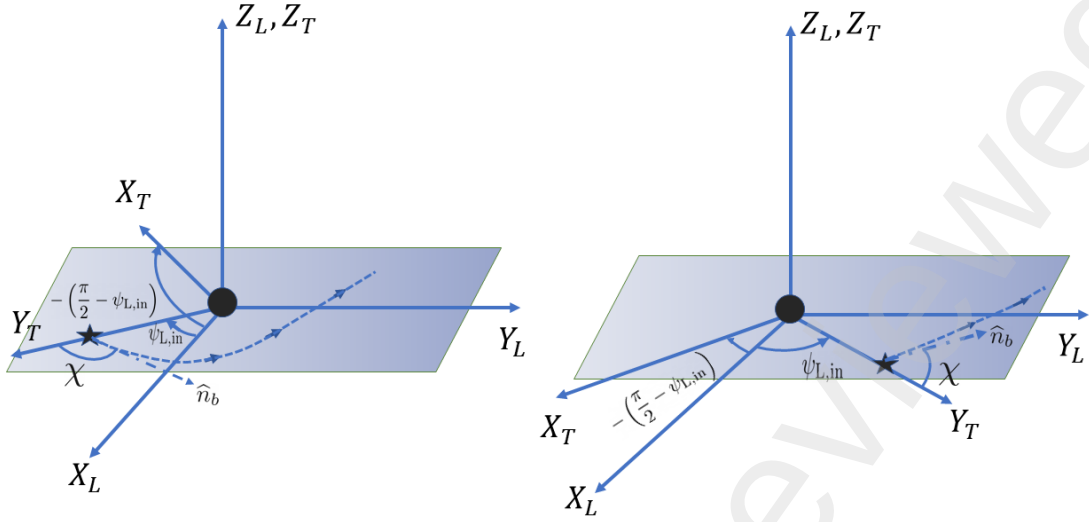


Figure 5: The orientation of the L-frame with respect to the T-frame. The blue plane is the $X_L Y_L$ plane. The left figure is for the case when $\psi_{L,in}$ is negative and the right figure is for the case when $\psi_{L,in}$ is positive. In both of the panels, the pulsar is indicated by a black star, and the companion is represented by a filled black circle.

origin, their Z-axis is also the same, i.e., $Z_T \equiv Z_L$. Hence, following Eq. (15), the unit vector along the Z_L -axis (in the b-frame) can be written as:

$$\widehat{z}_L = \widehat{r}_b \times \widehat{n}_b, \quad (20)$$

Although the X_L -axis and the Y_L axis lie in the $X_T Y_T$ plane, their directions are usually different from those of the X_T -axis and the Y_T -axis. For this, we denote the azimuthal angle of the light ray, i.e., the angle between the light ray and the X_L -axis as ψ_L .

The impact parameter D of the light ray is defined as the perpendicular distance of its initial direction from the companion. If one knows the initial direction of the light ray and the position of the pulsar at the time of emission, then D can be determined from the expression given by Poutanen (2020):

$$D = \frac{|\vec{r}_c| \sin \chi}{\sqrt{1 - \frac{GM_c}{|\vec{r}_c|^2}}}, \quad (21)$$

where M_c is the mass of the companion, G is the gravitational constant, χ is the angle between the pulsar's position vector in the c-frame, $\vec{r}_c$, and $\widehat{n}_b$ is the initial direction of the light ray in the b-frame, which is also the initial direction of the light ray in the c-frame (as the c-frame and the b-frame are parallel to each other). Fig. 4 shows D , χ , and $\widehat{n}_b$.

Using D from Eq.(21), and following the derivation of null geodesics from Chandrasekhar (1983), we can determine the initial azimuthal angle $\psi_{L,in}$ of the light ray in the L -frame, which is also the azimuthal angle of the source (e.g., the pulsar) in the L -frame at the time of emission. In the T -frame, we align the initial position of the source along the Y_T -axis. Hence, the angle between the X_L axis and the Y_T axis at the time of emission is $\psi_{L,in}$.

This makes the angle between the X_L -axis and the X_T -axis to be $|\pi/2 - \psi_{L,in}|$. More specifically, we define the angle as $+(\pi/2 - \psi_{L,in})$ when measured from the X_T -axis to the X_L -axis, and $-(\pi/2 - \psi_{L,in})$ when measured from the X_L -axis to the X_T -axis. As a result, a rotation of the X_T -axis by an angle $(\pi/2 - \psi_{L,in})$ about the Z_T -axis brings it into alignment with the X_L -axis (see Fig. 4). The Y_L -axis is then obtained using the following cross product:

$$\widehat{y}_L = \widehat{z}_L \times \widehat{x}_L. \quad (22)$$

where $\widehat{x}_L$, $\widehat{y}_L$, and $\widehat{z}_L$ are the unit vectors directed along the X_L , Y_L , and Z_L axes, respectively.

Note that, in the context of gravitational bending of the pulsar signal observed by a distant observer, only light rays with the impact parameter $D > 3\sqrt{3}GM_c/c^2$ are relevant (Chandrasekhar, 1983, c is the speed of light in vacuum). When such a light ray reaches closest to the companion, it is taken as the X_L -axis, i.e., $\psi_L|_{closest} = 0$ (as shown in Fig. 4).

Now, as the requirement of our definition, we need to align the Z -axis of the L -frame, Z_L , with the Z_T -axis. To achieve this, we have adopted the convention that when the light ray initially moves towards the gravitating body, $\psi_{L,in}$ is negative, and when it initially moves away from the companion, $\psi_{L,in}$ is positive. These two situations are depicted in the left and the right panels of Fig. 5, respectively.

The solution to the null geodesics yields the final azimuthal position of the light ray at infinity, denoted as $\psi_{L,\infty}$, which determines its asymptotic direction in the L -frame as

$$\widehat{n}_{L,\infty} = \begin{bmatrix} \cos \psi_{L,\infty} \\ \sin \psi_{L,\infty} \\ 0 \end{bmatrix}. \quad (23)$$

We can then transform this unit vector into any relevant coordinate frame using the transformation relations presented in the next section.

3. Transformation relations between different frames

In this section, we find the transformation relations between different frames and define the related angles to study the binary pulsar. To relate the frames already mentioned, sometimes we encounter some intermediate frames that we will see in due course. These intermediate frames are denoted with symbols prime ($\prime$) or double primes ($\prime\prime$). Two such primed frames, I' and sc' , are illustrated in Fig. 2; these are related to their unprimed counterparts by a translation along the barycentric position vector. In addition, we introduce several intermediate coordinate frames that arise during the rotational transformations between these frames, as we discuss next.

In the discussion that follows, we use the symbol $R_N^P(\theta)$ to denote a passive rotation about N -axis by an angle θ and $R_N^A(\theta)$ to denote an active rotation about N -axis by an angle θ , where N can be any of X, Y, Z .

3.1. Transformation from the m -frame to the I -frame

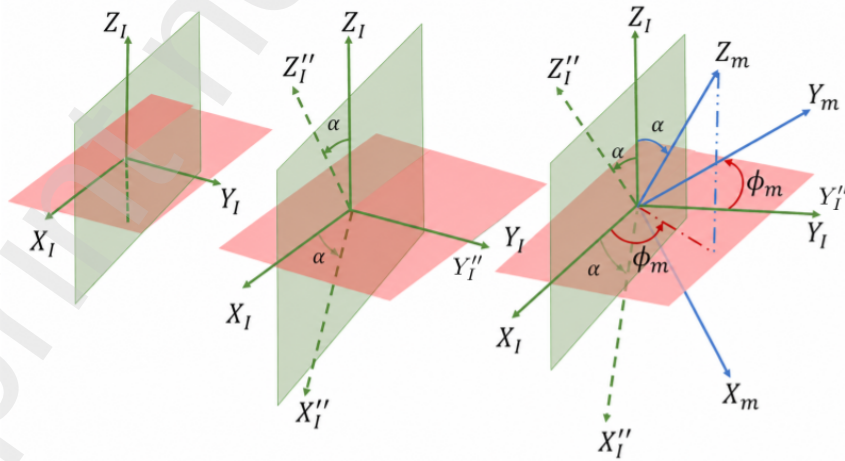


Figure 6: Transformation from the I -frame to the m -frame. The green plane is the $Z_I X_I$ plane, and the red plane is the $X_I Y_I$ plane.

From the definitions described in Subsection 2.1.1, we know that the angle between the Z_I -axis and the Z_m -axis is α and the projection of the Z_m -axis on the $X_I Y_I$ -plane makes an angle ϕ_m

with the X_I axis. The transformation from the I-frame to the m-frame can be done in two steps as described below.

- (i) By rotating the I-frame about the Y_I -axis by an angle α , we get a new frame $X_I'' Y_I'' Z_I''$. Here Y_I'' is along the Y_I -axis. The Z_I'' -axis and the X_I'' -axis are in the $Z_I X_I$ plane. The angle between the X_I -axis and the X_I'' -axis is the same as the angle between the Z_I -axis and the Z_I'' -axis, i.e., α . This rotation is shown in the middle panel of Fig. 6.
- (ii) Next, we rotate the $X_I'' Y_I'' Z_I''$ frame about the Z_I -axis by an angle ϕ_m to get the $X_m Y_m Z_m$ frame. As the rotation is about the Z_I -axis, Y_m remains in the $X_I Y_I$ plane (the red plane in Fig. 6). This rotation is shown in the right panel of Fig. 6.

Together, these transformations can be summarized as:

$$X_I Y_I Z_I \xrightarrow{R_{Y_I}^P(\alpha)} X_I'' Y_I'' Z_I'' \xrightarrow{R_{Z_I}^P(\phi_m)} X_m Y_m Z_m . \quad (24)$$

Hence, the reverse transformation can be written as:

$$X_m Y_m Z_m \xrightarrow{R_{Z_I}^P(-\phi_m)} X_I'' Y_I'' Z_I'' \xrightarrow{R_{Y_I}^P(-\alpha)} X_I Y_I Z_I . \quad (25)$$

Thus, if a vector is denoted by $\vec{V}_m$ in the m-frame and by $\vec{V}_I$ in the I-frame, these two would be related as

$$\vec{V}_I = R_Z^A(\phi_m) R_Y^A(\alpha) \vec{V}_m . \quad (26)$$

3.2. Transformation from the I-frame to the I'-frame

As described in Subsection 2.2.5, the I-frame (origin at the center of the companion) and the I'-frame (origin at the center of the pulsar) are related by a translation without any rotation, any vector in these two frames is essentially the same, i.e., if a vector in the I-frame be denoted by $\vec{V}_I$ and the same vector in the I'-frame by $\vec{V}_{I'}$ we can write,

$$\vec{V}_I = \vec{V}_{I'} . \quad (27)$$

3.3. Transformation from the I'-frame to the s-frame

In Subsection 2.2.2, we discussed how the orientation of the spin axis of the pulsar, i.e., the $Z_{I'}$ -axis is specified with respect to the pS-frame. As the I'-frame is obtained from the I-frame with a translation without any rotation (Subsection 3.4), the Z_I -axis and the $Z_{I'}$ -axis have the same orientation. Moreover, as the pS-frame is obtained from the s-frame with a translation without any rotation, the orientation of the $Z_{I'}$ -axis in the s-frame is the same as the orientation of the Z_I -axis in the pS-frame.

Hence, following Subsection 2.2.2, the angle between the Z_s -axis and the $Z_{I'}$ -axis is denoted by λ_p , and the angle between the X_s -axis and the projection of the $Z_{I'}$ axis on the $X_s Y_s$ -plane is denoted by η_p . Using this information, it is easier to visualize the transformation of the s-frame to the I'-frame in two steps.

1. A rotation of the s-frame about the Y_s -axis by an angle λ_p gives a new frame $X_s' Y_s' Z_s'$ where Y_s' is along the Y_s -axis, i.e., the Y_s' -axis is in the $X_s Y_s$ plane. The Z_s' -axis and the X_s' -axis are in the $Z_s X_s$ plane. The angle between the Z_s -axis and the Z_s' -axis as well as the angle between the X_s -axis and the X_s' -axis is λ_p . This transformation is shown in the middle panel of Fig. 7.
2. We rotate the $X_s' Y_s' Z_s'$ frame about the Z_s -axis by an angle η_p to get $X_{I'} Y_{I'} Z_{I'}$ frame. As the rotation is about the Z_s -axis, the $Y_{I'}$ -axis remains in the $X_s Y_s$ plane. This transformation is shown in the middle panel of Fig. 7.

Together, the above two-step transformation can be represented as:

$$X_s Y_s Z_s \xrightarrow{R_{Y_s}^P(\lambda_p)} X_s' Y_s' Z_s' \xrightarrow{R_{Z_s}^P(\eta_p)} X_{I'} Y_{I'} Z_{I'} . \quad (28)$$

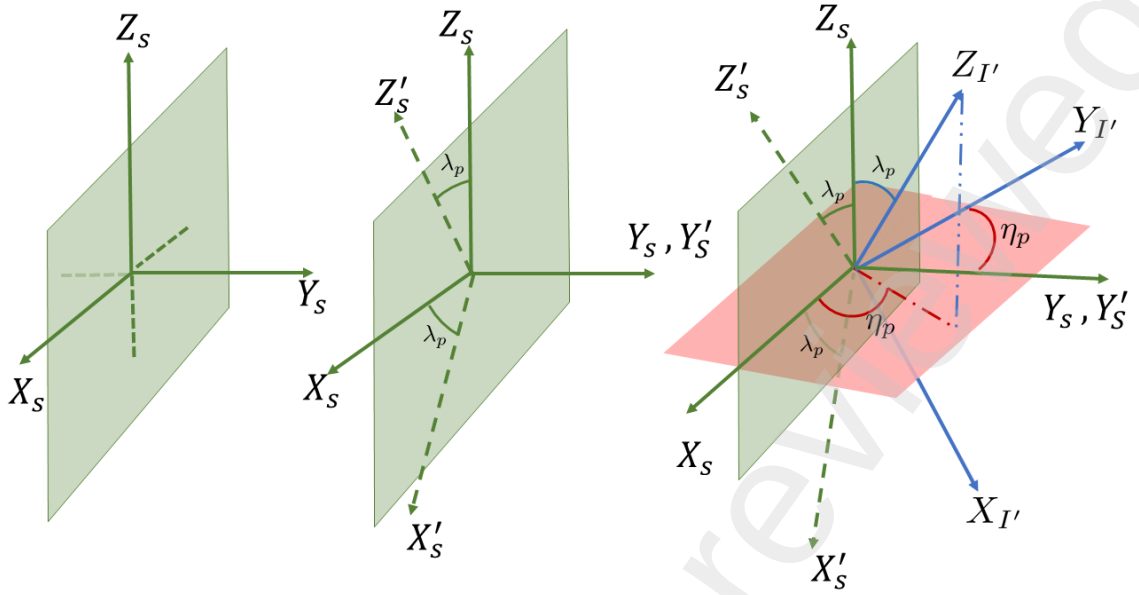


Figure 7: The transformation from the s-frame to the I-frame. The green plane is the $Z_s X_s$ plane and the red plane is the $X_s Y_s$ plane.

Hence, the reverse transformation can be written as:

$$X_{I'} Y_{I'} Z_{I'} \xrightarrow{R_{Z_s}^p(-\eta_p)} X'_s Y'_s Z'_s \xrightarrow{R_{Y'_s}^p(-\lambda_p)} X_s Y_s Z_s . \quad (29)$$

Thus, if a vector is denoted by $\vec{V}_I$ in the I-frame and by $\vec{V}_s$ in the s-frame, these two would be related by

$$\vec{V}_s = R_Z^A(\eta_p) R_Y^A(\lambda_p) \vec{V}_{I'} . \quad (30)$$

3.4. Transformation from the I-frame to the s-frame

As we mentioned in Subsection 3.2, the I-frame and the I'-frame are identical, the above transformations relate the I-frame to the s-frame as:

$$X_s Y_s Z_s \xrightarrow{R_{Y_s}^p(\lambda_p) R_{Z_s}^p(\eta_p)} X_I Y_I Z_I , \quad (31)$$

and the reverse transformation as

$$X_I Y_I Z_I \xrightarrow{R_{Z_s}^p(-\eta_p) R_{Y_s}^p(-\lambda_p)} X_s Y_s Z_s . \quad (32)$$

Thus, if a vector is denoted by $\vec{V}_I$ in the I-frame and by $\vec{V}_s$ in the s-frame, these two would be related by

$$\vec{V}_s = R_Z^A(\eta_p) R_Y^A(\lambda_p) \vec{V}_I . \quad (33)$$

3.5. Transformation from the s-frame to the b-frame

We have mentioned in Subsection 2.2.3, that the angle between Z_s -axis and Z_b is denoted by i and the angle between the X_s -axis and the X_b -axis is denoted by ω . Hence, the b-frame can be obtained from the s-frame through the following two steps:

1. A rotation of the s-frame about the Z_s -axis by an angle ω gives a new frame $X''_s Y''_s Z''_s$ where Z''_s is along the Z_s -axis and the X''_s -axis and the Y''_s -axis are in the $X_s Y_s$ plane. This transformation is shown in the middle panel of Fig. 8. The angle between the X_s -axis and the X''_s -axis as well as the angle between the Y_s -axis and the Y''_s -axis is ω .

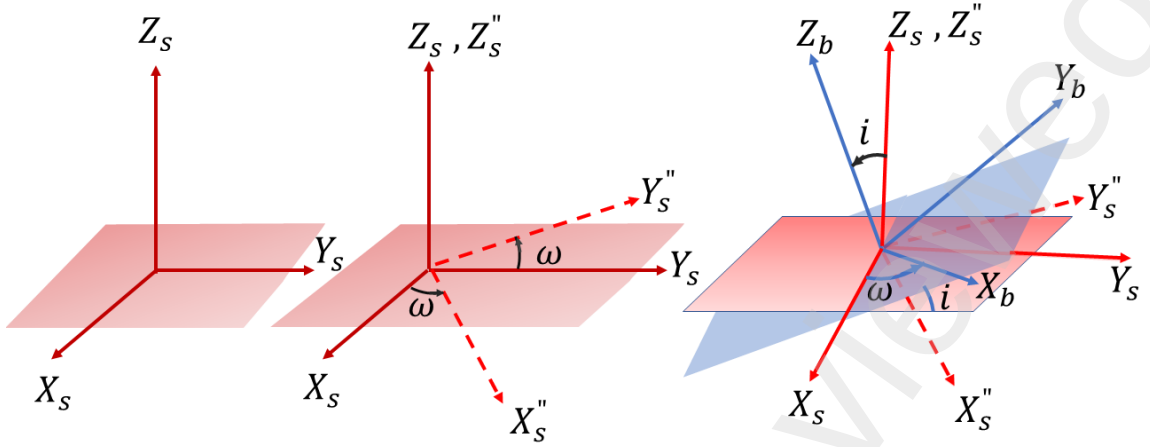


Figure 8: The transformation from the s-frame to the b-frame. The red plane is the $X_s Y_s$ plane (the sky plane), and the blue plane is the $X_b Y_b$ plane (the orbital plane). The X_s -axis is the line of the nodes, which is along the intersection of the sky-plane and the orbital plane.

2. Now, a rotation of the $X''_s Y''_s Z''_s$ frame about the X''_s -axis by an angle i gives the $X_b Y_b Z_b$ frame. This transformation is shown in the right panel of Fig. 8, which ensures that the angle between the X''_s -axis and the X_b -axis, between the Y''_s -axis and the Y_b -axis, as well as that between the Z''_s -axis and the Z_b -axis are i .

Together, the above two-step transformation can be represented as:

$$X_s Y_s Z_s \xrightarrow{R_{Z_s}^p(\omega)} X''_s Y''_s Z''_s \xrightarrow{R_{X''_s}^p(i)} X_b Y_b Z_b . \quad (34)$$

Thus, if a vector is denoted by $\vec{V}_s$ in the s-frame and by $\vec{V}_b$ in the b-frame, these two would be related by

$$\vec{V}_b = R_Z^A(-\omega) R_X^A(-i) \vec{V}_s . \quad (35)$$

or,

$$\vec{V}_s = R_X^A(i) R_Z^A(\omega) \vec{V}_b . \quad (36)$$

Note that, instead of Eq. (34), if the second transformation is a rotation of the $X''_s Y''_s Z''_s$ frame about the Y''_s -axis by an angle i , we will still get a new frame whose Z-axis would make an angle i with the Z_s -axis, but the X_s -axis will not remain along the intersection of the sky-plane (the $X_s Y_s$ plane) and the $X Y$ plane of the new frame, i.e., the X_s -axis will not be the line of node, if we assume the $X Y$ plane of the new frame as the b-frame. This situation is contrary to our definition of the X_s -axis being along the line of node.

3.6. Transformation from the I-frame to the b-frame

To express vectors in the b-frame, we proceed in two steps:

1. Transform the vector from the I-frame to the s-frame as described in Subsection 3.4.
2. Transform the result from the s-frame to the b-frame as described in Subsection 3.5.

Combining Eqs. (35) and (30), we get,

$$\vec{V}_b = R_Z^A(-\omega) R_X^A(-i) \vec{V}_s = R_Z^A(-\omega) R_X^A(-i) R_Z^A(\eta_p) R_Y^A(\lambda_p) \vec{V}_I . \quad (37)$$

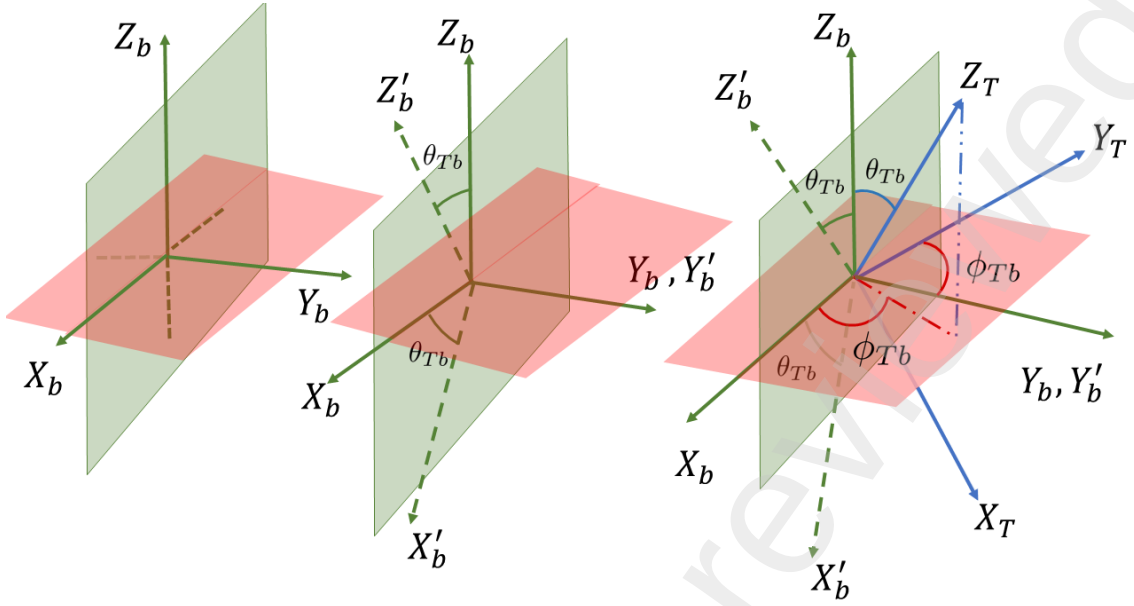


Figure 9: The transformation from the b-frame to the T-frame. The green plane is the $Z_b X_b$ plane and the red plane is the $X_b Y_b$ plane or the orbital plane.

3.7. Transformation from the T-frame to the b-frame

In Subsection 2.2.8, we defined θ_{Tb} as the angle between the Z_T -axis and the Z_b -axis and ϕ_{Tb} as the angle between the Y_T -axis and the Y_b -axis. Then the transformation from the b-frame to the T-frame can be done in two steps as:

1. A rotation of the b-frame about the Y_b -axis by an angle θ_{Tb} gives a new frame $X'_b Y'_b Z'_b$. This transformation is shown in the middle panel of Fig. 9. Here, the Y'_b -axis is along the Y_b -axis and both of the Z'_b -axis and the X'_b -axis are in the $Z_b X_b$ plane. The angle between the X_b -axis and the X'_b -axis is the same as the angle between the Z_b -axis and the Z'_b -axis, i.e., θ_{Tb} .
2. Next, a rotation of the $X'_b Y'_b Z'_b$ frame about the Z_b -axis by an angle ϕ_{Tb} gives the $X_T Y_T Z_T$ frame. This transformation is shown in the right panel of Fig. 9.

The above two transformations can be written as:

$$X_b Y_b Z_b \xrightarrow{R_{Y_b}^P(\theta_{Tb})} X'_b Y'_b Z'_b \xrightarrow{R_{Z_b}^P(\phi_{Tb})} X_T Y_T Z_T. \quad (38)$$

Hence, the reverse transformation can be written as:

$$X_T Y_T Z_T \xrightarrow{R_{Z_b}^P(-\phi_{Tb})} X'_b Y'_b Z'_b \xrightarrow{R_{Y_b}^P(-\theta_{Tb})} X_b Y_b Z_b. \quad (39)$$

Thus, if a vector is denoted by $\vec{V}_T$ in the T-frame and by $\vec{V}_b$ in the b-frame, these two would be related by

$$\vec{V}_b = R_Z^A(\phi_{Tb}) R_Y^A(\theta_{Tb}) \vec{V}_T. \quad (40)$$

Note that, if to the b-frame, one first applies a rotation about the X_b -axis by an angle θ_{Tb} followed by a rotation about the Z_b -axis by an angle ϕ_{Tb} , then also one would obtain a new frame whose Z -axis would make an angle θ_{Tb} with the Z_b -axis and the X -axis would make an angle ϕ_{Tb} with the X_b -axis. However, in that case, the Y -axis of the new frame would not be in the orbital plane, which would contradict our choice of the Y_T -axis being along $\hat{r}_b$.

3.8. Transformation from the sc-frame to the sc'-frame

The sc-frame is defined with its origin at the center of the spinning companion. We shift this origin to the barycenter keeping the directions of the axes unchanged. This gives a new frame called the sc'-frame.

As described in Subsection 2.2.6, that the sc-frame (origin at the center of the companion) and the sc'-frame (origin at the center of the companion) are related by a translation without any rotation, any vector in these two frames is basically the same, i.e., if a vector in the sc-frame be denoted by $\vec{V}_{sc}$ and the same vector in the sc'-frame by $\vec{V}_{sc'}$, we can write,

$$\vec{V}_{sc} = \vec{V}_{sc'}. \quad (41)$$

3.9. Transformation from the the sc'-frame to the s-frame

The transformation from the sc'-frame to the s-frame can be understood from the definition of the sc'-frame as described in Subsection 2.2.6. This can be done in two steps as described below.

1. A rotation of the s-frame about the Y_s -axis by an angle λ_{sc} gives a new frame $X'_s Y'_s Z'_s$ where the Y'_s -axis is along the Y_s -axis, i.e., the Y'_s -axis is still in the sky plane (the $X_s Y_s$ plane). This transformation is shown in the middle panel of Fig. 10. Here, we see that the Z'_s -axis and the X'_s -axis are in the $Z_s X_s$ plane. The angle between the Z_s -axis and the Z'_s -axis as well as the angle between the X_s -axis and the X'_s -axis is λ_{sc} .
2. Next, a rotation of the $X'_s Y'_s Z'_s$ frame about the Z'_s -axis by an angle η_{sc} gives the $X'_{sc} Y'_{sc} Z'_{sc}$ frame. This transformation is shown in the right panel of Fig. 10. As the rotation is about the Z'_s -axis, Y'_{sc} still remains in the $X_s Y_s$ plane, i.e., in the sky-plane.

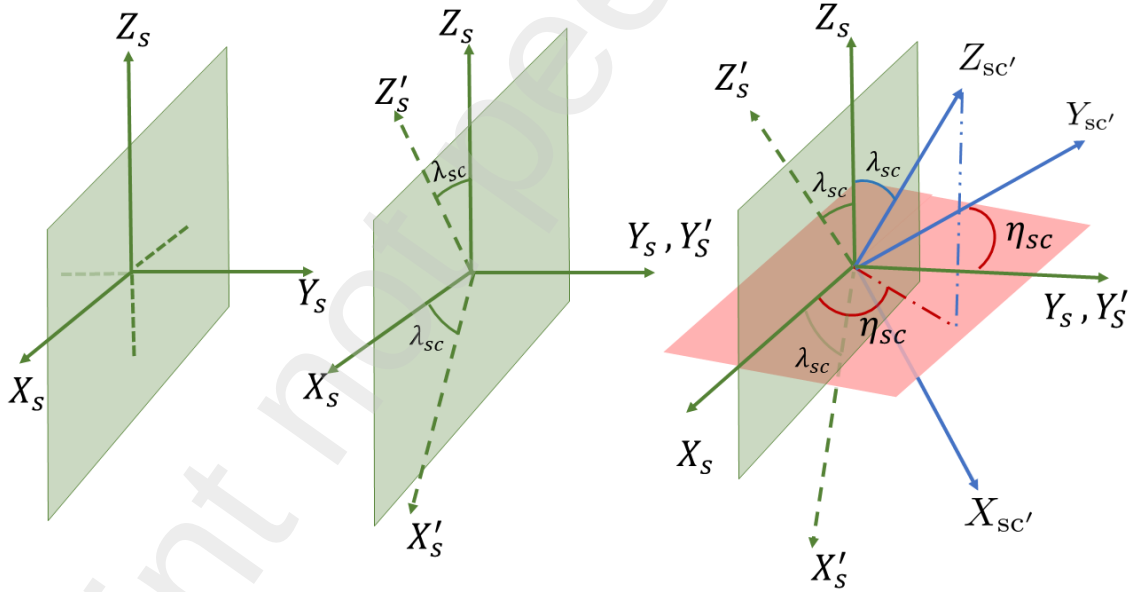


Figure 10: The transformation from the s-frame to the sc'-frame. The green plane is the $Z_s X_s$ plane and the red plane is the $X_s Y_s$ plane.

The above set of transformations can be summarized as:

$$X_s Y_s Z_s \xrightarrow{R_{Y_s}^P(\lambda_{sc})} X'_s Y'_s Z'_s \xrightarrow{R_{Z'_s}^P(\eta_{sc})} X'_{sc} Y'_{sc} Z'_{sc}. \quad (42)$$

Hence, the reverse transformation can be written as:

$$X'_{sc} Y'_{sc} Z'_{sc} \xrightarrow{R_{Z'_s}^P(-\eta_{sc})} X'_s Y'_s Z'_s \xrightarrow{R_{Y_s}^P(-\lambda_{sc})} X_s Y_s Z_s. \quad (43)$$

Thus, if a vector is denoted by $\vec{V}_{sc'}$ in the sc'-frame and by $\vec{V}_s$ in the s-frame, then these two would be related as follows:

$$\vec{V}_s = R_Z^A(\eta_{sc}) R_Y^A(\lambda_{sc}) \vec{V}_{sc'}. \quad (44)$$

3.10. Relating the forms of a vector in the sc-frame and in the b-frame

Using the right hand side of Eq. (44) in the left hand side of Eq. (36), we get,

$$R_Z^A(\eta_{sc}) R_Y^A(\lambda_{sc}) \vec{V}_{sc'} = R_X^A(i) R_Z^A(\omega) \vec{V}_b. \quad (45)$$

Hence, the relation between the forms of any given vector in the b-frame ($\vec{V}_b$) and in the sc'-frame ($\vec{V}_{sc'}$) is given by:

$$\vec{V}_{sc'} = R_Y^A(-\lambda_{sc}) R_Z^A(-\eta_{sc}) R_X^A(i) R_Z^A(\omega) \vec{V}_b. \quad (46)$$

Eq. (41) implies that the relation between the forms of any given vector in the b-frame ($\vec{V}_b$) and in the sc-frame ($\vec{V}_{sc}$) is the same, i.e.,

$$\vec{V}_{sc} = R_Y^A(-\lambda_{sc}) R_Z^A(-\eta_{sc}) R_X^A(i) R_Z^A(\omega) \vec{V}_b. \quad (47)$$

3.11. Relating the forms of a vector in the sc-frame and in the T-frame: Transformation from the sc'-frame to the T-frame

Using the right hand side of Eq. (40) into the right hand side of Eq. (47), we get the relation between the forms of any given vector in the sc'-frame ($\vec{V}_{sc'}$) and in the T-frame ($\vec{V}_T$) as:

$$\vec{V}_{sc'} = R_Y^A(-\lambda_{sc}) R_Z^A(-\eta_{sc}) R_X^A(i) R_Z^A(\omega) R_Z^A(\phi_{Tb}) R_Y^A(\theta_{Tb}) \vec{V}_T. \quad (48)$$

Eq. (41) implies that the relation between the forms of any given vector in the sc-frame ($\vec{V}_{sc}$) and in the T-frame ($\vec{V}_T$) is the same, i.e.,

$$\vec{V}_{sc} = R_Y^A(-\lambda_{sc}) R_Z^A(-\eta_{sc}) R_X^A(i) R_Z^A(\omega) R_Z^A(\phi_{Tb}) R_Y^A(\theta_{Tb}) \vec{V}_T. \quad (49)$$

3.12. Transformation from the L-frame to the T-frame

We have defined the T-frame in Subsection 2.2.8 and the L-frame in Subsection 2.2.9 where we have seen that the X_T axis makes an angle $-(\pi/2 - \psi_{L,in})$ with the X_L axis (see Fig. 4). Hence, the T-frame can be obtained from the L-frame by rotating it about the Z_L -axis through as:

$$X_L Y_L Z_L \xrightarrow{R_{Z_L}^p(-(\pi/2 - \psi_{L,in}))} X_T Y_T Z_T. \quad (50)$$

Thus, if a vector is denoted by $\vec{V}_L$ in the L-frame and by $\vec{V}_T$ in the T-frame, the two representations are related by

$$\vec{V}_T = R_Z^A(\pi/2 - \psi_{L,in}) \vec{V}_L. \quad (51)$$

4. Discussions

In this short review, we have synthesized and explicitly defined the primary coordinate frames commonly used to study astrophysical phenomena around pulsars, along with the mathematical relations required to transform between them. We hope these explicit definitions will serve as a clear, practical reference for analytical studies across pulsar physics - whether applying post-Newtonian approximations in binary systems, modeling the general relativistic effects around a strongly gravitating object (like a black hole), or mapping the local magnetospheric geometry of pulsars.

As the Square Kilometre Array Observatory (SKAO) telescopes come online, high sensitivity of the array and its broad radio frequency coverage will transform pulsar astronomy. For instance, the discovery of exotic pulsars, particularly near the Galactic center (Abbate et al., 2025), will offer unprecedented opportunities to test gravity in strong-field regimes (Venkatraman Krishnan et al., 2025). Interpreting such systems requires precise frame conversions between the observer's sky plane, orbital axes, and pulsar I-frame.

Similarly, Pulsar Timing Array (PTA) science with the SKAO telescopes aims to detect nanohertz gravitational waves and conduct fundamental tests of physics (Shannon et al., 2025).

Consistent coordinate definitions across local pulsar frames, Solar System barycentric coordinates, and observational frames are essential to eliminate alignment errors in high-precision timing models. Furthermore, the unprecedented sensitivity of the SKAO telescopes will allow us to probe emission mechanics and magnetospheric structures in much greater detail (Oswald et al., 2025), the modeling of which requires seamless transformations between the corotating magnetic dipole m-frame and the pulsar I-frame.

Finally, future discoveries and higher observational precision will likely reveal subtle relativistic features, such as how light bending in curved spacetime affects pulse polarization profiles (Debnath et al., in prep.). Correctly interpreting these signatures requires robust analytical models grounded in clear coordinate frame transformations. By establishing a unified convention for these reference frames, we hope this review provides a reliable mathematical foundation for the analytical study and modeling of pulsar physics.

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